

Multiple Choice

1. **Answer:** D

Options: A) Silicon; B) Diamond; C) Silicon carbide; D) Arsenic-doped silicon

Note: Correct option: D - Arsenic-doped silicon

2. **Answer:** D

Options: A) $6.345 \times 10^{-4} \text{ at } 0.335T$; $9.871 \times 10^{-5} \text{ at } 10.5T$; B) $0.793 \times 10^{-4} \text{ at } 0.335T$; $6.931 \times 10^{-7} \text{ at } 10.5T$; C)

Note: Correct option : D – $4.126 \times 10^{-3} \text{ at } 0.335T$; $2.142 \times 10^{-6} \text{ at } 10.5T$

2. **Answer:** D

Options: A) Melting point; B) Boiling point; C) Polarizability; D) First ionization energy

Note: Correct option: D - First ionization energy

3. **Answer:** D

Options: A) F; B) Si; C) O; D) Ca

Note: Correct option: D - Ca

4. **Answer:** A

Options: A) HCl; B) AgCl; C) CaCl_2 ; D) CCl_4

Note: Correct option: A - HCl

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '3.74 T'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 1:8:28:56:70:56:28:8:1. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
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Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.